

KRITAGYA TIMSINA

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PROFESSIONAL SUMMARY

Purpose-driven Computer Engineering student with strong interest in full-stack web development, AI-powered applications, and scalable software systems. Experienced in building modern web applications using **Next.js, React, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, and Python**. Proven ability to create responsive user-focused products, integrate authentication and payment systems, and deliver AI-assisted applications with real-time functionality. Committed to leveraging software and machine learning to solve practical problems through clean architecture, efficient backend systems, and intuitive user experiences.

EDUCATION

Khwopa College of Engineering

Libali Bhaktapur, Nepal

Bachelor in Computer Engineering (BCT), **Tribhuvan University Affiliated**

Expected Graduation: May 2026

PROJECTS

AI Dental Assistant App (github.com/kritagyatimsina023/dentelligent.git)

Personal Project

- Engineered a full-stack AI-driven dental assistance platform supporting voice AI agent interactions across 4+ core dental query categories.
- Architected appointment booking, doctor availability management, conflict handling, and real-time status updates, eliminating manual scheduling overhead across all user flows.
- Integrated secure authentication and user management with Clerk, covering end-to-end payment workflow for appointment handling across 2+ user roles.
- Structured scalable frontend and backend using Next.js, TanStack Query, Zustand, Prisma ORM, PostgreSQL, Clerk, and VAPI across 5+ integrated services.

Emotion-Based Symbolic Music Generation (github.com/kritagyatimsina023/aiMusicBackend.git)

Major Project

- Spearheaded an AI-driven music generation system processing lyrics and captions to produce emotion-aligned music output across 3 distinct generation stages.
- Leveraged the Valence-Arousal model to classify lyrical mood into multiple emotional dimensions, mapping 2 psychological axes to improve music-caption alignment accuracy.
- Introduced a 3-stage generation pipeline producing distinct Stage-1, Stage-2, and GAN-based Stage-3 variations, offering users 3x more creative output options per session.
- Launched the full-stack application using React.js, TypeScript, Firebase Auth, Node.js, Express.js, MongoDB, Flask, and Python-based ML routes spanning 8+ technologies.

Real-Time Chat Application (github.com/kritagyatimsina023/chatter.git)

Personal Project

- Engineered a real-time messaging platform enabling instant user-to-user communication with synchronized chat history retrieval for unlimited concurrent sessions.
- Secured all user sessions using JWT-based authentication, protecting 100% of API endpoints from unauthorized access.
- Integrated Socket.io achieving sub-second message delivery and live notification handling across all active user connections.
- Consolidated 7 technologies — React.js, Zustand, Axios, Node.js, Express.js, MongoDB, and Socket.io — into a cohesive full-stack system with unified state management.

Query AI (github.com/kritagyatimsina023/nextGPT.git)

Personal Project

- Crafted an AI-driven conversational web application delivering ChatGPT-style responses via Gemini 2.5 Flash API integration.
- Accelerated response generation by connecting directly to Gemini 2.5 Flash API, enabling real-time AI output with minimal latency across all query types.

- Expanded monetization coverage by adding 2 payment gateways — **eSewa** (regional) and Stripe (global) — supporting users across multiple billing regions.
- Unified frontend and backend workflows using Next.js, MongoDB, AI API integration, and payment gateway systems into a single deployable application.

Spylt Milk Clone Website (github.com/kritagyatimsina023/spylt-milk.git) Frontend Project

- Conceived and produced a highly interactive, responsive clone website centered on modern product presentation and animation-driven UX across all major screen sizes.
- Executed 5+ advanced scroll-based and text-based animations using GSAP ScrollTrigger and SplitText, reducing average user bounce indicators through visual engagement.
- Built a performant frontend using Next.js, GSAP, and responsive design principles optimized for cross-device compatibility.

MovieMate (github.com/kritagyatimsina023/movie-mate.git) Frontend Project

- Developed a movie information platform enabling users to search and explore details for thousands of current titles using optimized rendering for every search query.
- Streamlined external data access by integrating the TMDB API, delivering live movie metadata with a fully responsive UI across all device breakpoints.
- Composed a lightweight, state-managed frontend using React.js, Zustand, and TMDB API integration with under 3 core dependencies for state handling.

Code Book (github.com/kritagyatimsina023/codebook.git) Personal Project

- Established an e-commerce themed online book purchasing application with end-to-end shopping functionality covering 3+ core modules: browsing, cart, and checkout.
- Secured all routes using JWT-based authentication and protected routing, covering 100% of user-facing pages from unauthorized access.
- Implemented using React.js, Redux, React Router DOM, JSON Server, and frontend authentication workflows with centralized state management across all application layers.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS, C/C++

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, PyTorch, Pandas, NumPy, Zustand, TanStack Query, Redux

Databases & ORM: MongoDB, PostgreSQL, MySQL, Prisma ORM

Tools & Platforms: Git, GitHub, Firebase Auth, Clerk, Socket.io, RESTful API, VAPI, Stripe, eSewa

Soft Skills: Team Collaboration, Agile Development